

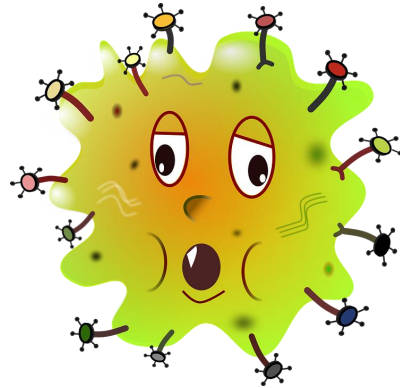
BRONCHIOLITIS

The background features a dark gradient with numerous out-of-focus red and white circular lights (bokeh) scattered across the frame. A dense, horizontal trail of fine red particles or dust extends from the left side towards the center, adding a sense of motion or depth to the composition.

What is bronchiolitis?

- is a common lower respiratory tract **infection** that affects babies and young children under 2 years old. It causes **inflammation** and congestion in the small airways (**bronchioles**) of the lung.

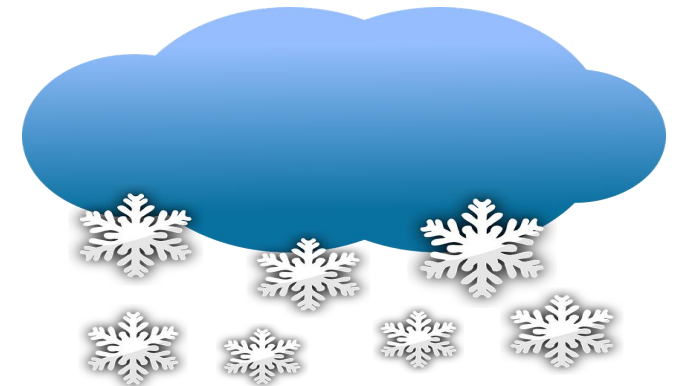
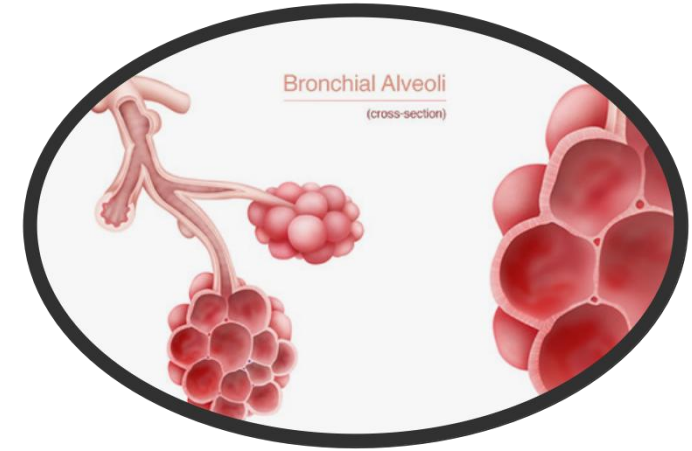
- It is caused by **VIRUS**



- It is common in babies less than 1 year old or neonates

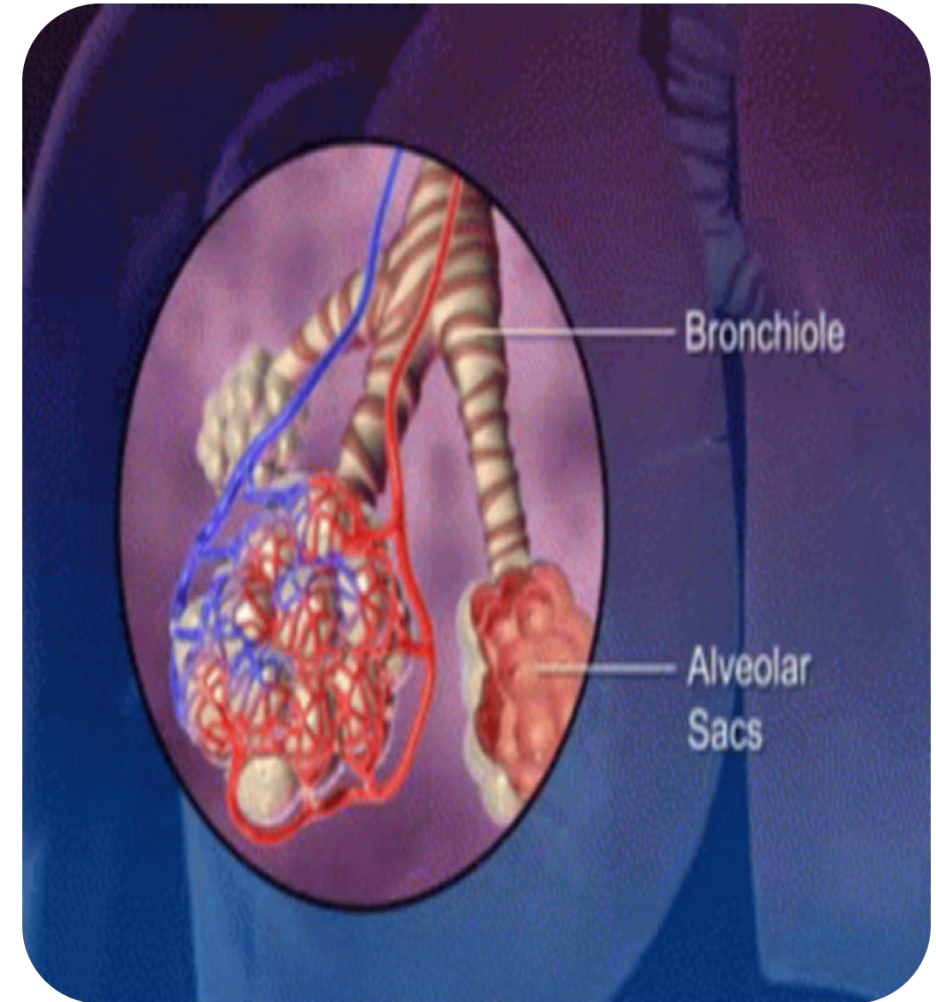


- The peak time for bronchiolitis is during the **winter** months or cold season

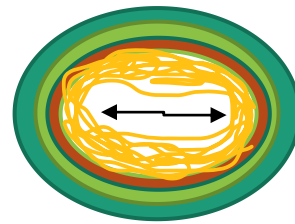


Structure of bronchioles

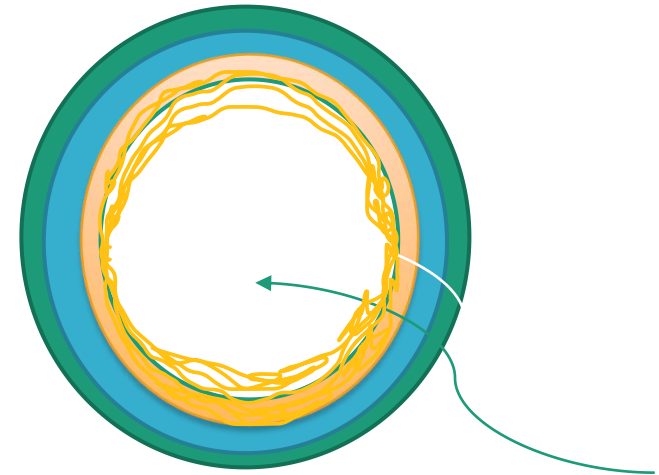
- are air passages or small airways inside the lungs that branch off like tree limbs from the bronchi which lead to the alveoli where gas exchange occurs
- bronchioles are made of smooth muscle layers to facilitate *bronchoconstriction* and *bronchodilatation*
- the bronchial lining is made up ciliated columnar epithelium which is characterized by presence of columnar cells covered with cilia and scattered mucus secreting **goblet cells**
- In the distal part of the bronchial airway, located are a group of nonciliated bronchiolar secretory cells known as *clara cells* that behave as a kind of bronchial surfactant – limiting lung collapse.



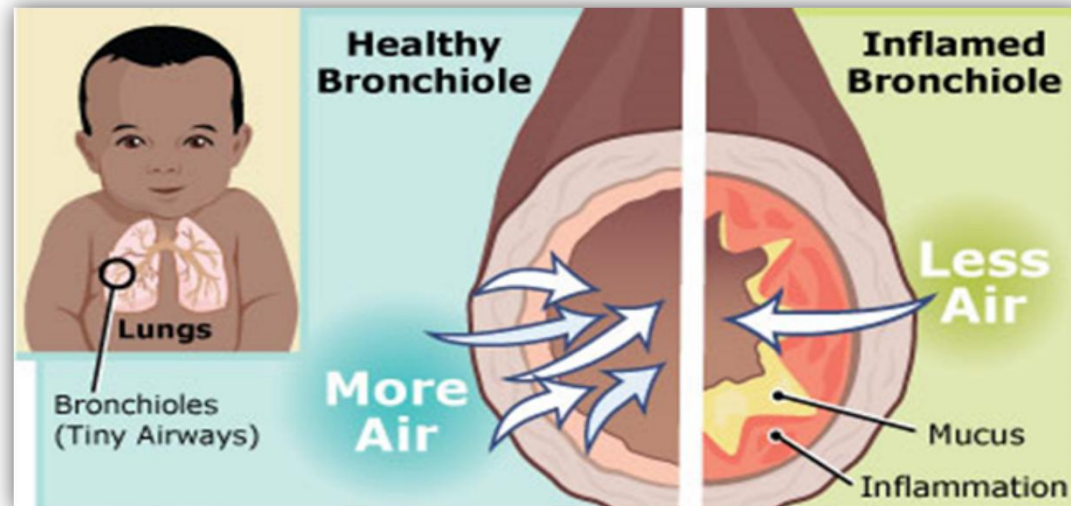
INFANTS



ADULTS

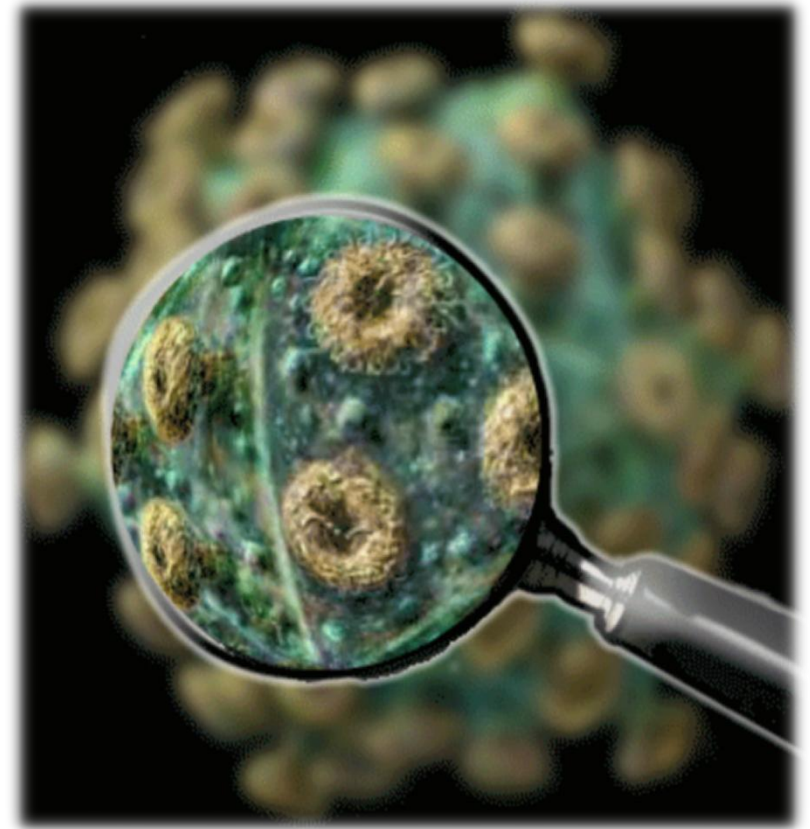


Air

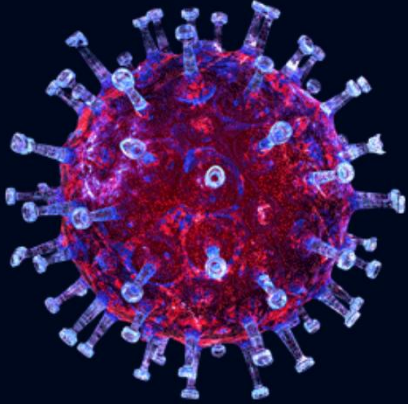


What causes bronchiolitis?

- Respiratory Syncytial virus (RSV)
- Rhinovirus
- Parainfluenza
- Adenovirus
- Influenza Virus
- Corona virus



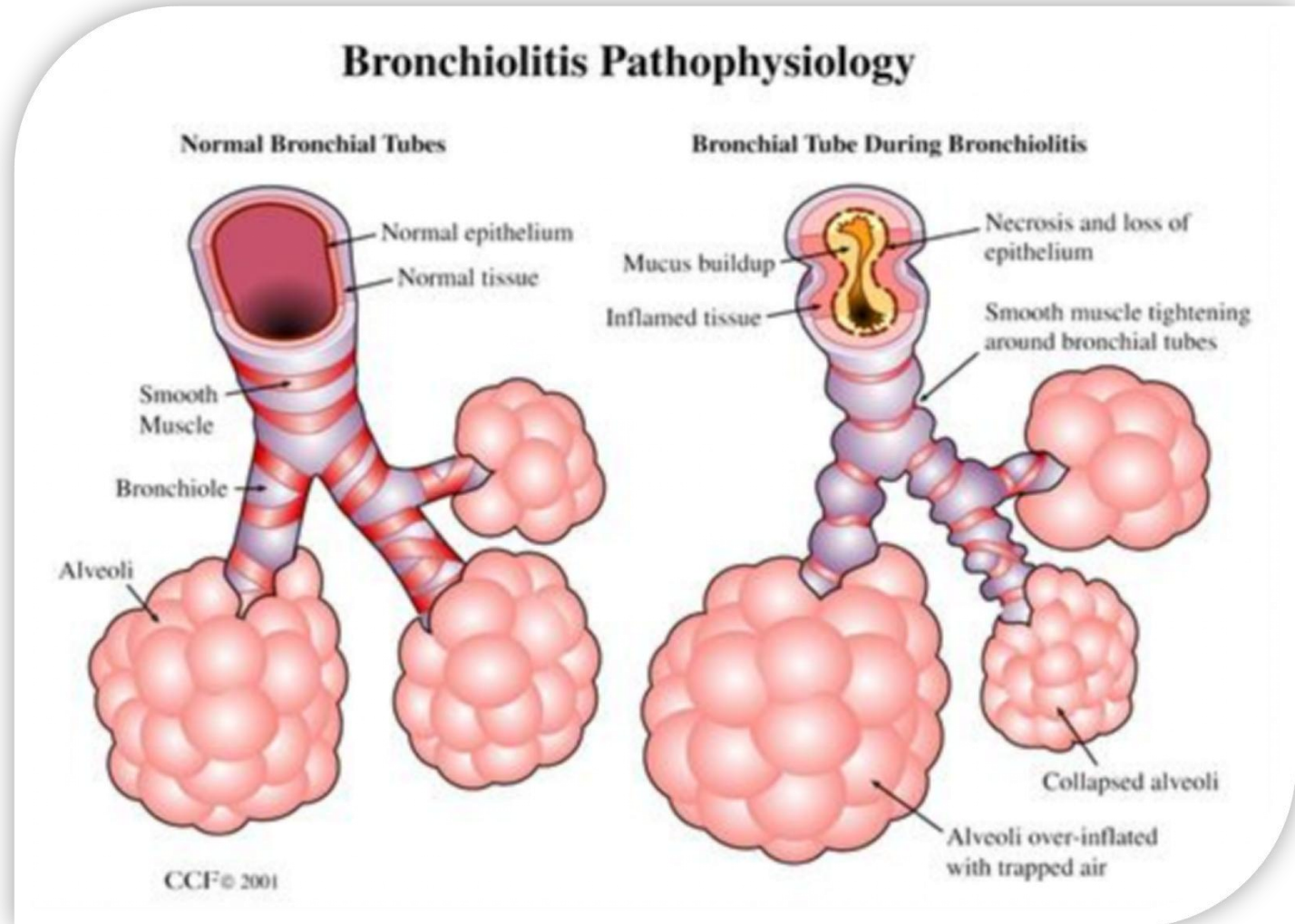
Respiratory Syncytial Virus



- is the most common cause of bronchiolitis
- ssRNA virus
- airborne droplets/direct contact

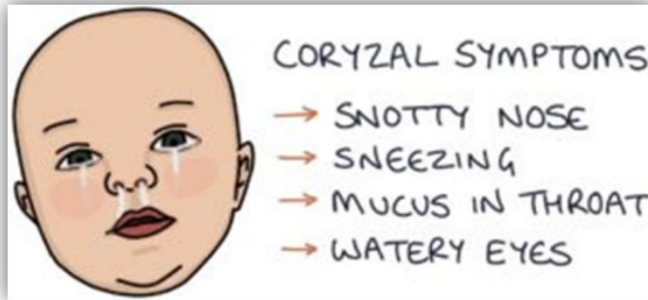
- ❖ Necrosis of the respiratory epithelium is one of the earliest lesions of bronchiolitis
- ❖ Proliferation of goblet cells results in excessive mucus production
- ❖ Epithelial regeneration with nonciliated cells impairs elimination of secretions
- ❖ Lymphocytic infiltrations may result in submucosal edema
- ❖ The pathology results in obstruction of bronchioles by *inflammation*, *edema* and *debris* leading to:

- hyperinflation
- increased airway resistance
- atelectasis



Presentation of bronchiolitis

CORYZAL SYMPTOMS



rapid or noisy breathing (wheeze)

vomiting after feeding

mild fever not more than 39°C



rapid or persistent cough

brief pauses in breathing

Bronchiolitis severity classification:

MILD

- **active and alert**
- **feeding or taking orally well**
- **none to minimal intercostal retractions**
- **respiratory rate within normal**

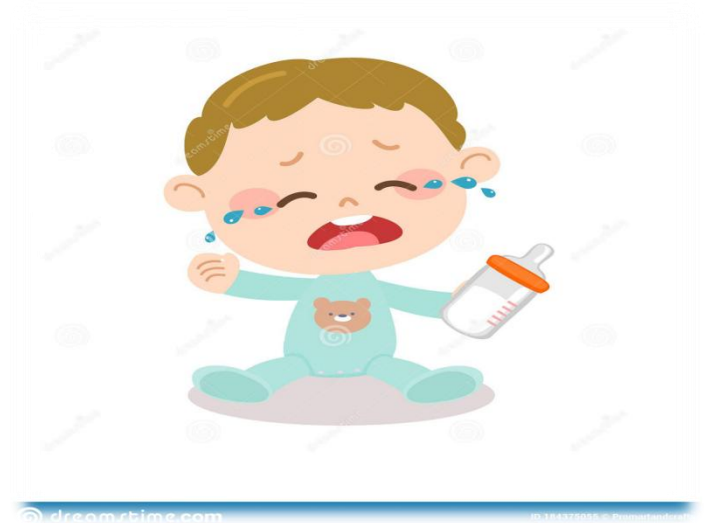
MODERATE

- alert and able to be consoled
- slight decrease in feeding or oral intake
- tachypnea
- low grade fever
- nasal flaring
- minimal to moderate intercostal recession
- inspiratory crepitation
- abdominal breathing
- tracheal tugging

Auscultation:
*wheeze

SEVERE

- high respiratory rate or bradypnea
- cyanosis
- apnea
- severe grunting
- difficulty in feeding less than 50% usual consumption



Who are at risk?

- under 1 year old or infants; common < 6 months of age
- immunocompromised individuals
- Premature babies
- congenital heart disease
- chronic lung disease



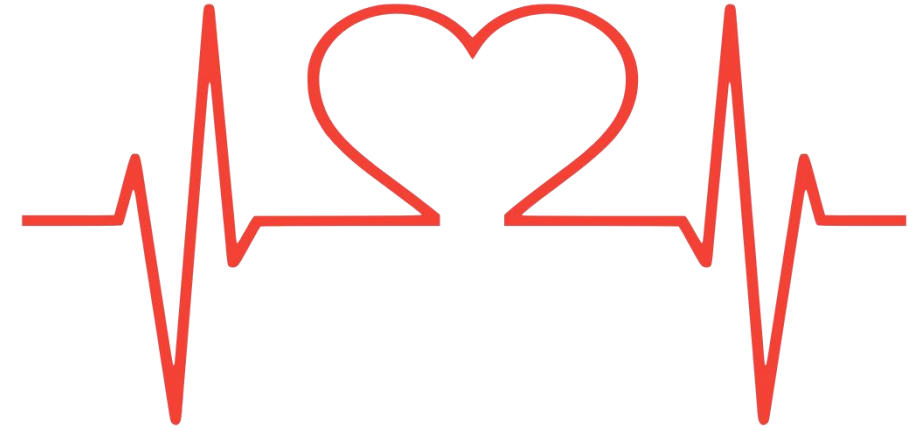
* Rarely occurs up to 2 years

Criteria for admission

- **< 3 months old**
- **premature babies**
- **< 50%-75% of normal intake of milk**
- **Clinically dehydrated**
- **Cyanosis**
- **Apnea**
- **Desaturations, < 92% oxygenation**



CLINICAL DIAGNOSIS



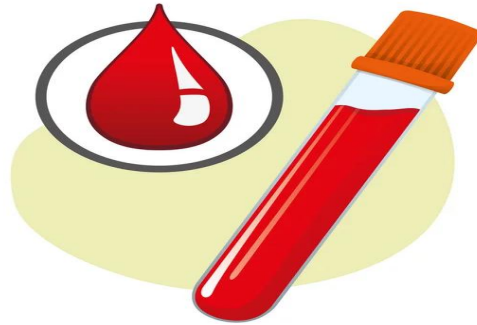
- **BRONCHIAL ASTHMA**
- **PNEUMONIA**
- **WHOOPING COUGH**
- **CONGESTIVE HEART FAILURE**
- **FOREIGN BODY ASPIRATION**

How is Bronchiolitis diagnosed?

- **Viral testing or swab**



- **Blood test**



- **Imaging tests:
chest x-ray or chest CT scan**



Treatment of bronchiolitis

HOME CARE

- **Ensure adequate intake or make sure that child drinks enough and well hydrated**
- **Saline drops or nasal suctioning**
- **Keep child on upright position**
- **Humidify the air**
- **Consider over-the-counter pain relievers or anti-pyretic for fever**

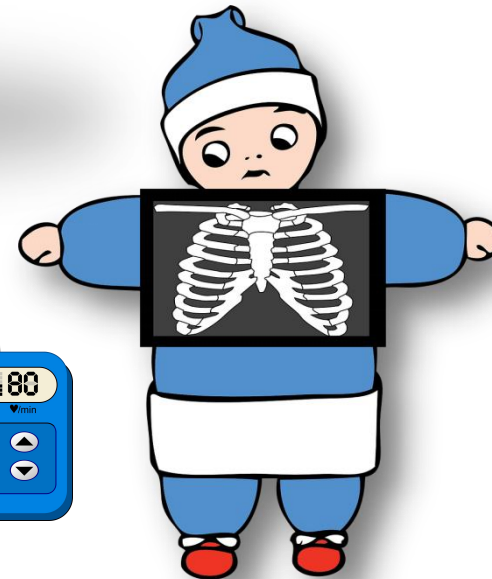
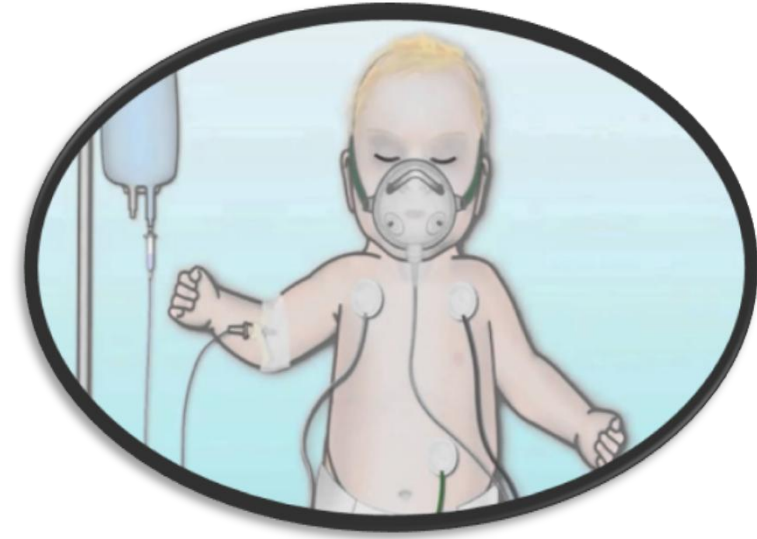


Hospital care

- **Administer supplemental humidified OXYGEN**
- **Ensure adequate intake or good hydration status; fluid replacement**
 - oral feeds
 - NGT feeds
 - IV fluids
- **Ensure patent airway; perform nasal and oral suctioning**
- **Monitor for apnea**
- *** Ventilatory support: high flow, NIV and intubation**

Investigations:

- **Swab testing**
- **Xray**
- **Electrolytes**
- **Blood cultures >38.5°C fever**
- **Pulse oximetry**
- **Blood gases**



Prevention



- Wash hands or sanitize hands
- Avoid close contact with sick people
- Clean and disinfect surfaces
- Cover mouth during coughs and sneezes
- Breastfeed
- Be vaccinated

Thank
You!

